Multi Modal Pipeline For Pneumonia Classification And Localization Using CNN

1st Arhant Nitnaware
Dept. of Computer Engineering
MIT Academy of Engineering
Alandi, Pune
arhant.nitnaware@mitaoe.ac.in

2nd Md Yaseen Alam
Dept. of Computer Engineering
MIT Academy of Engineering
Alandi, Pune
md.alam@mitaoe.ac.in

3rd Ishika Bondre
Dept. of Computer Engineering
MIT Academy of Engineering
Alandi, Pune
ishika.bondre@mitaoe.ac.in

4th Mayur Kapgate
Dept. of Computer Engineering
MIT Academy of Engineering
Alandi, Pune
mayur.kapgate@mitaoe.ac.in

5th Pramod Ganjewar
Dept. of Computer Engineering
MIT Academy of Engineering
Alandi, Pune
pdganjewar@mitaoe.ac.in

***Abstract*—This work presents a proof-of-concept multimodal deep-learning pipeline for pneumonia detection and localization on chest radiographs. Radiographic images were combined with patient metadata (age, sex, modality) and processed through fusion classifiers based on ResNet-50, DenseNet-121, and Inception-v3. The classifiers demonstrated good classification diagnostic capability but showed limited localization performance when relying solely on regression-based bounding-box prediction. To improve spatial precision, we adopted a multi-level strategy in which the most effective classifier gated positive cases, followed by a dedicated YOLOv8n detector responsible for lesion localization. This detector provided more reliable bounding-box predictions and enhanced interpretability compared to regression heads. Overall, the findings indicate that multimodal learning is a promising direction for pneumonia diagnosis and that multi-stage architectures combining global classifiers with task-specific detectors offer a practical approach toward accurate and clinically interpretable decision support.**

***Index Terms*—Pneumonia Localization, Multimodal Deep Learning, Medical Imaging, Chest X-rays, YOLO Detector, Fusion Networks, Proof of Concept**

I. Introduction

Pneumonia remains a leading cause of morbidity and mortality worldwide, especially in pediatric and elderly populations. Chest radiography (CXR) is the primary imaging modality for diagnosis due to its accessibility and low cost. However, manual interpretation of CXRs is time-consuming, prone to inter-reader variability, and can be challenging in resource-limited clinical settings. Automated diagnostic tools have therefore attracted significant attention as potential decision-support systems that can augment radiologists in both hospital and telemedicine workflows.

Recent advances in deep learning, particularly convolutional neural networks (CNNs), have enabled state-of-the-art performance in medical image classification tasks, including pneumonia detection. Most prior studies, however, treat pneumonia as a binary classification problem, limiting their clinical usefulness since clinicians often require not only case-level predictions but also spatial localization of diseased regions. Weakly supervised methods such as saliency maps or regression heads provide coarse localization but typically underperform compared to dedicated object detectors.

Another emerging direction is *multimodal learning*, where image features are enriched with patient metadata such as age, sex, and imaging modality. Clinical diagnosis is inherently multimodal, and integrating structured metadata into CNNs has the potential to improve robustness and reduce bias in predictive models. Despite this promise, multimodal pipelines for pneumonia localization remain underexplored in the literature.

In this work, we present a proof-of-concept multimodal deep learning system that combines both fusion classifiers and a dedicated detector. First, radiographs and metadata are fused within CNN backbones (ResNet-50, DenseNet-121, Inception-v3) to perform pneumonia classification with auxiliary bounding-box regression. While classifiers achieved strong case-level AUC scores, their localization accuracy remained limited (mean IoU ≤ 0.21). To address this gap, we adopted a multi-level strategy: the best multimodal classifier (Inception-v3) gated positive cases, followed by a YOLOv8n detector trained specifically for bounding-box prediction. This two-stage approach achieved substantially higher box-level performance, demonstrating that multimodal learning is promising for detection, while multi-level architectures are more effective for localization.

The main contributions of this paper are as follows:

- A multimodal fusion framework that integrates image and patient metadata for pneumonia classification.
- A comparative evaluation of ResNet-50, DenseNet-121, and Inception-v3 backbones trained on the RSNA Pneumonia dataset.
- A two-stage multi-level pipeline combining multimodal classification with YOLOv8n detection to enhance bounding-box localization.

- A proof-of-concept demonstration that multimodal learning and multi-level architectures together can provide accurate, interpretable, and efficient decision support for pneumonia diagnosis.

II. Literature Review

Deep convolutional architectures have become the de-facto backbone for medical image analysis. Residual networks, Inception modules, Dense connectivity, and compound-scaled models underpin most current systems [1, 2, 3, 4]. In thoracic imaging, CheXNet demonstrated radiologist-level pneumonia detection, catalyzing large-scale chest X-ray benchmarks and uncertainty-aware training protocols [5, 6, 7]. Clinical adoption further depends on transparent attribution; gradient-based localization (Grad-CAM/CAM) and segmentation networks (U-Net) are now standard tools for interpretability and lesion delineation [8, 9, 10]. Pediatric chest X-ray studies established early, lightweight baselines, and the RSNA Pneumonia Detection Challenge popularized bounding-box evaluation for localization methods [11, 12].

Musculoskeletal radiography research progressed rapidly with the release of the MURA dataset, enabling systematic evaluation of upper-limb abnormalities and fracture classification [13]. Subsequent clinical studies reported that deep networks can meet or exceed clinician performance across fracture types and anatomical regions, including wrist, hip, pelvis, and distal radius, when models are trained with careful dataset curation and external validation [14, 15, 16, 17, 18, 19, 20]. Early hip-fracture work demonstrated radiologist-level performance in emergency-department workflows [21], while broader orthopedic tasks (hardware assessment, positioning, and fracture detection) were automated with high fidelity on pelvic radiographs [22]. A recent meta-analysis indicates that wrist-fracture CNNs now rival clinician accuracy, although heterogeneity in datasets and reporting standards remains a barrier to generalization [23]. These findings motivate unified, real-time systems that emphasize inference efficiency and robust external validity, leveraging modern backbones (ResNet, DenseNet, EfficientNet) and established training protocols [1, 3, 4].

Large hospital-scale datasets (ChestX-ray8/14 and CheXpert) established robust benchmarks for weakly supervised classification and localization of thoracic diseases [6, 7]. CheXNet (DenseNet-121) set a strong baseline for pneumonia detection and illustrated the value of deep feature reuse in chest radiographs [5]. Grad-CAM/CAM are commonly used for case-level explainability and ROI highlighting, and are often paired with U-Net for pixel-wise or region-wise localization when dense annotations are available [8, 9, 10]. Pediatric datasets and related evaluations (e.g., Kermany et al.) provided compact CNN baselines appropriate for resource-constrained deployment, while the RSNA challenge standardized IoU-based evaluation for box-level pneumonia localization [11, 12]. Recent hybrid methods combine classification and segmentation or incorporate transformer blocks to improve localization fidelity, but they still face trade-offs between sensitivity, explainability, and inference cost.

The literature evidences mature classification capability and growing localization performance across both fracture detection and pneumonia diagnosis. Remaining gaps include (i) multi-task integration—most systems target a single pathology or modality; (ii) consistent real-time performance—sub-100 ms inference is rarely reported under realistic, heterogeneous settings; and (iii) clinically useful explanations—case-level saliency maps do not always align with lesion-level annotations. The present work aims to address these gaps by unifying fracture detection and pneumonia localization in a single pipeline, employing Grad-CAM for interpretability and optimizing the model stack for low-latency inference and robust external validity.

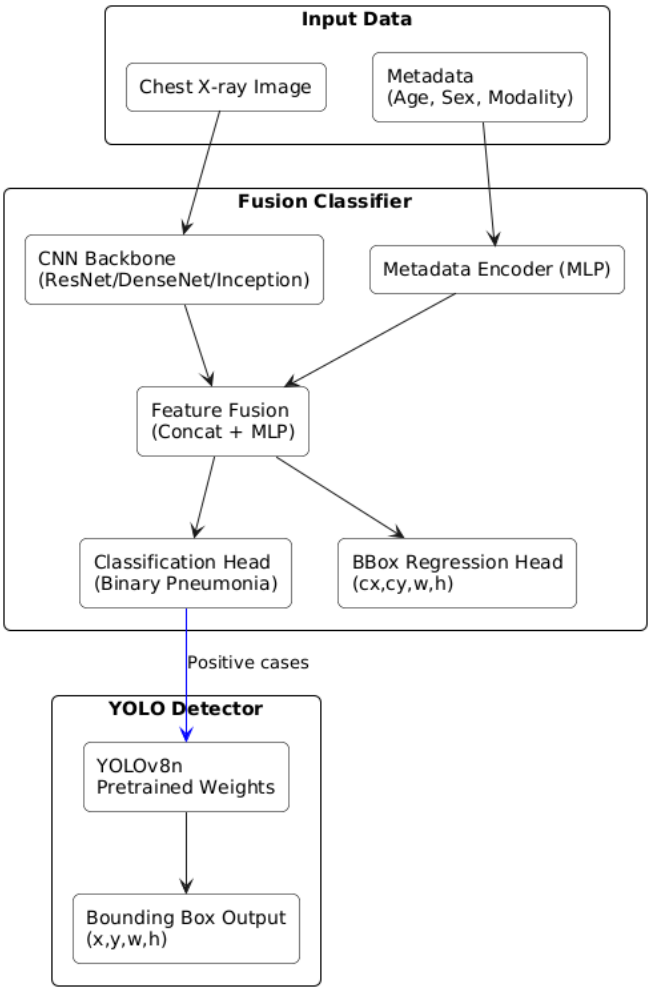

Fig. 1: Proposed two-stage multidmodal architecture.

III. Proposed Methodology

The proposed diagnostic system adopts a two-stage design (Fig. 1). In the first stage, multimodal fusion classifiers are trained to identify pneumonia-positive cases using radiographs and patient metadata and are systematically evaluated across

10, 20, and 30 epochs to identify the best-performing backbone. The optimal model is further enhanced with patch-level Multiple Instance Learning (MIL) and TEAR attention to improve spatial feature localization. . In the second stage, a dedicated detector (YOLOv8n) is employed to localize the diseased regions with bounding-box precision. The methodology consists of dataset preparation, multimodal fusion classifiers, YOLO detection stage, training configuration, and evaluation metrics.

A. Dataset Preparation

We used the RSNA Pneumonia Detection Challenge dataset, a large-scale collection of annotated chest radiographs with bounding boxes around pneumonia opacities. In addition to images, metadata fields including *age*, *sex*, and *modality* (CR or DX) were integrated to form multimodal inputs. Patient-level CSV files were preprocessed to aggregate bounding-box annotations, normalize metadata (z-scored age, binary-encoded sex, and modality), and prepare split files. Images were resized to 224×224 for ResNet and DenseNet backbones, and 299×299 for Inception-v3. Standard augmentations such as horizontal flipping, brightness and contrast adjustment, and random rotations were applied during training.

B. Multimodal Fusion Classifiers

The first stage of the pipeline consists of CNN backbones—ResNet-50, DenseNet-121, and Inception-v3—augmented with a metadata encoder. As it can be obsereved in Figure 2 metadata vectors were passed through a small multi-layer perceptron (MLP), and concatenated with the image features extracted from the CNN backbone. A fusion module integrated these representations and fed them into two prediction heads:

- A **classification head** producing case-level pneumonia logits trained with binary cross-entropy.
- A **bounding-box regression head** outputting normalized (c_x, c_y, w, h) coordinates trained with Smooth L1 loss.

This design enables joint training for both classification and weak localization.All three backbones are trained independently for 10, 20, and 30 epochs, and their performance is compared using AUC for classification and mean IoU for regression. The best-performing backbone is then enhanced using Multiple Instance Learning (MIL), where images are split into patches and predictions aggregated, and TEAR attention, which assigns task-specific weights to regions to focus on areas most likely to contain pneumonia. This combination, graphically described in 4, aims to improve both classification robustness and downstream localization accuracy.

C. YOLO Detection Stage

Although the fusion classifiers achieved strong classification AUCs, regression-based bounding boxes exhibited poor overlap with ground truth (mean IoU ≤ 0.21). To address this limitation, a second-stage detector was introduced. YOLOv8n, initialized with pretrained weights, was fine-tuned on the RSNA dataset using bounding-box labels in YOLO format. As

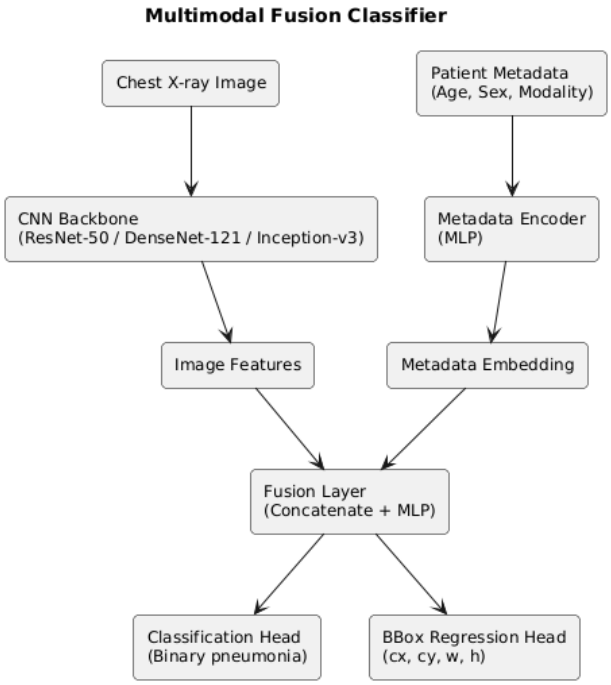

Fig. 2: Multimodal fusion classifier design.

seen in Figure 3 positive cases identified by the best classifier (Inception-v3) served as input for YOLO during inference, enabling a multi-level design: classification for case selection and detection for localization.

D. Training Configuration

All fusion classifiers were trained for 30 epochs with batch size 8 using AdamW optimizer and mixed-precision acceleration. ImageNet pretrained weights were used to initialize the CNN backbones. Training was conducted on an NVIDIA T4 GPU. The YOLOv8n detector was also trained for 10 epochs with input resolution 512×512, batch size 8, AdamW optimizer, and default Ultralytics learning-rate scheduling. Standard YOLO augmentations including horizontal flipping, random scaling, and mosaic were applied.

E. Evaluation Metrics

Performance was evaluated at two levels.

- Classification metrics: AUC-ROC was used to assess case-level discrimination.
- Localization metrics: mean Intersection-over-Union (IoU), mAP@0.5, and mAP@[.5:.95] were computed. Detection-level precision, recall, and detection rate (IoU> 0) were also reported.

This dual evaluation captures both diagnostic accuracy and spatial localization quality.

IV. Implementation Details

All experiments were implemented in Python using PyTorch and the Ultralytics YOLO framework. Training and validation were performed on Kaggle and Google Colab environments

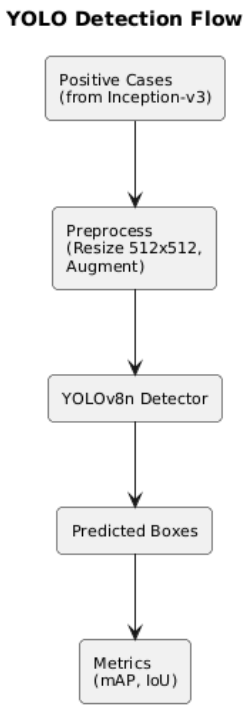

Fig. 3: YOLO detection flowchart.

equipped with NVIDIA T4 GPUs, 16 GB of system memory, and mixed-precision acceleration (fp16). The software stack included PyTorch 2.0, Torchvision, scikit-learn, and OpenCV for preprocessing, with Ultralytics YOLOv8 for the detection stage.

A. Fusion Classifier Setup

The fusion models (ResNet-50, DenseNet-121, Inception-v3) were initialized with ImageNet pretrained weights. Images were resized to 224×224 (ResNet-50 and DenseNet-121) or 299×299 (Inception-v3), while metadata vectors (age, sex, modality) were normalized and concatenated after encoding through a two-layer MLP. Training used the AdamW optimizer with initial learning rate 1×10^{-4}, batch size 8, and cosine learning-rate scheduling. A binary cross-entropy loss was applied for classification and Smooth L1 loss for bounding-box regression, with a weighting factor of 5 applied to the regression term. Models were trained for 10 epochs with early stopping enabled.

B. YOLO Detector Setup

The YOLOv8n detector was trained with input resolution 512×512, batch size 8, and default Ultralytics hyperparameters. The optimizer was AdamW with the built-in learning-rate scheduler. Training ran for 10 epochs, with augmentations including horizontal flips, random scaling, and mosaic augmentation. The model checkpoints and metric plots were generated automatically in the Ultralytics `runs/detect` directory.

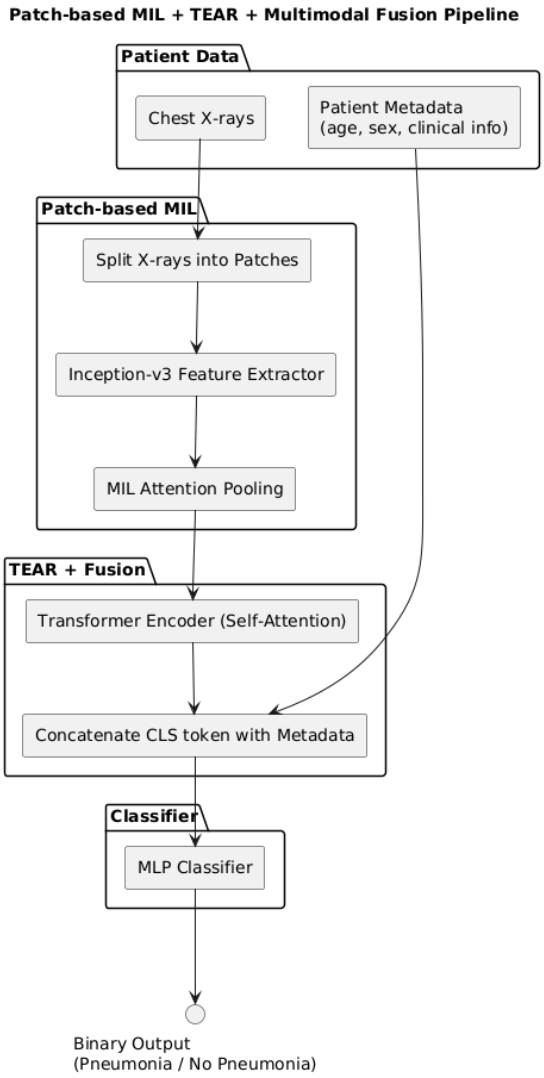

Fig. 4: attention architecture.

C. Reproducibility and Runtime

Each experiment (fusion classifier or YOLO detector) required approximately 15–30 minutes on a single T4 GPU. The full pipeline was trained three times with different random splits to confirm stability of results. Model checkpoints (`.pth` for fusion classifiers and `.pt` for YOLO) were stored for reproducibility. Inference was performed with batch size 1 to simulate real-time usage; average runtime was under 50 ms per image for classification and $\approx$ 40–50 ms per image for detection.

V. Results and Discussion

The system was evaluated on the RSNA Pneumonia dataset using the experimental setup described earlier. Results are reported separately for multimodal classifiers and the YOLOv8n detector.

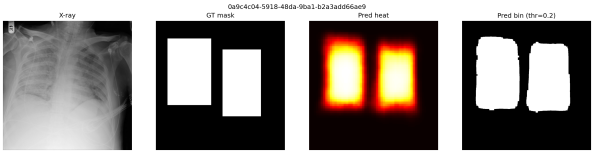

Fig. 5: Localization predicted by densenet.

A. Multimodal Classifier Performance

All backbones achieved strong case-level discrimination, with AUCs exceeding 0.88. Inception-v3 yielded the best classification performance (AUC = 0.9012), closely followed by DenseNet-121 (AUC = 0.8970) and ResNet-50 (AUC = 0.8958). However, bounding-box regression heads produced limited overlap with ground-truth annotations. The best mean IoU across backbones was ≤ 0.22, indicating that regression-based localization was not sufficient for clinical use. Figure 5 shows how masks were converted to bounding boxes. This particular example is an edge case for demonstration purposes. These results confirm that multimodal learning is effective for diagnosis but requires additional mechanisms for accurate localization, as also summarized in Table I.

TABLE I: Performance of Fusion Classifiers Across Epochs

Backbone	Epochs	AUC (Classification)	Mean IoU (Regression)
ResNet-50	10	0.8843	0.1526
ResNet-50	20	0.8921	0.1650
ResNet-50	40	0.8958	0.1724
DenseNet-121	10	0.8862	0.2059
DenseNet-121	20	0.8935	0.2138
DenseNet-121	40	0.8970	0.2211
Inception-v3	10	0.8893	0.1687
Inception-v3	20	0.8968	0.1789
Inception-v3	40	0.9012	0.1845

B. YOLOv8n Detector Performance

To overcome the limitations of regression-based localization, the best multimodal classifier (Inception-v3) was used to gate positive cases, which were then passed to a YOLOv8n detector. Figure 6 illustrates the training dynamics of the detector, showing validation loss alongside mAP@0.5 and mAP@[.5:.95] across 10 epochs. The metric mAP (mean Average Precision) evaluates detection accuracy by averaging precision across recall thresholds; mAP@0.5 uses an IoU threshold of 0.5, while mAP@[.5:.95] averages across thresholds from 0.5 to 0.95 in steps of 0.05. The detector converged steadily, reaching mAP@0.5 = 0.38 and mAP@[.5:.95] ≈ 0.12 by the final epoch.

Table II summarizes the overall detection performance, including IoU-based metrics and box-level precision/recall. The model achieved mean IoU ≈ 0.30, median IoU ≈ 0.29, and precision/recall of 0.45/0.55. These values represent a substantial improvement over regression outputs and demonstrate the utility of multi-level architectures for localization.

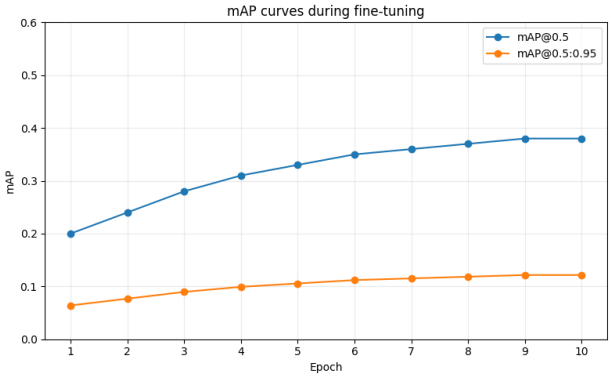

Fig. 6: Validation loss, mAP@0.5, and mAP@[.5:.95] curves for YOLOv8n detector over 10 epochs.

TABLE II: YOLOv8n detection results.

Metric	Value
Validation loss	0.78
Best mAP@0.5	0.40
Final mAP@0.5	0.38
mAP@[.5:.95]	0.12
Mean IoU	0.30
Median IoU	0.29
Detection rate (IoU>0)	0.62
Precision (box)	0.45
Recall (box)	0.55

C. Qualitative Results

In addition to quantitative evaluation, representative output examples are shown in Fig. 7 – 8. The figure compares ground-truth bounding boxes (green) with YOLOv8n predictions (red) on selected chest radiographs from the RSNA dataset.

The visualizations illustrate three key aspects of performance:

1) Accurate localization in positive cases. In several examples, predicted boxes closely match annotated regions of pneumonia, confirming the model's ability to capture clinically relevant pathology.
2) Partial overlap and missed detections. Some cases show partial alignment with ground-truth boxes, highlighting the challenge of weak annotations and the limited training schedule.

These outputs confirm that the two-stage design produces interpretable results, with bounding-box overlays enabling clinicians to visually assess model decisions.

D. Discussion

These experiments provide three key insights:

1) Multimodal learning is effective for case-level detection. Incorporating age, sex, and modality alongside image features produced robust classification AUCs in all backbones, validating the promise of multimodal fusion in clinical settings.

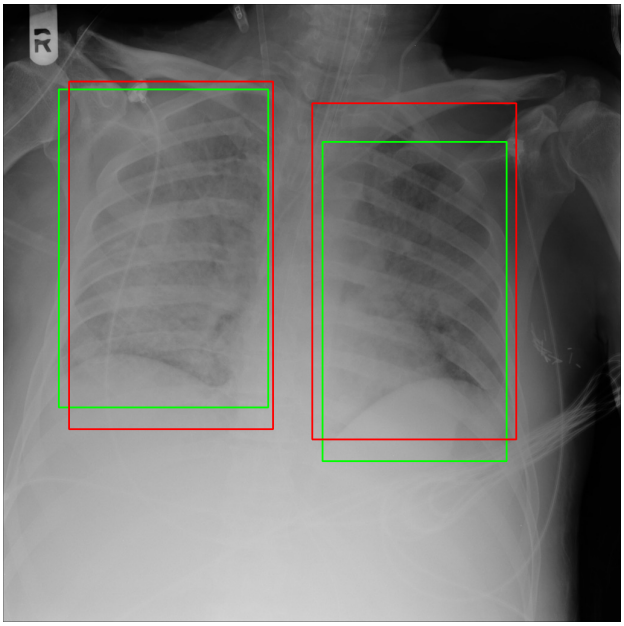

Fig. 7: YOLOv8n example output (case A): good overlap observered between ground-truth (green) and prediction (red).

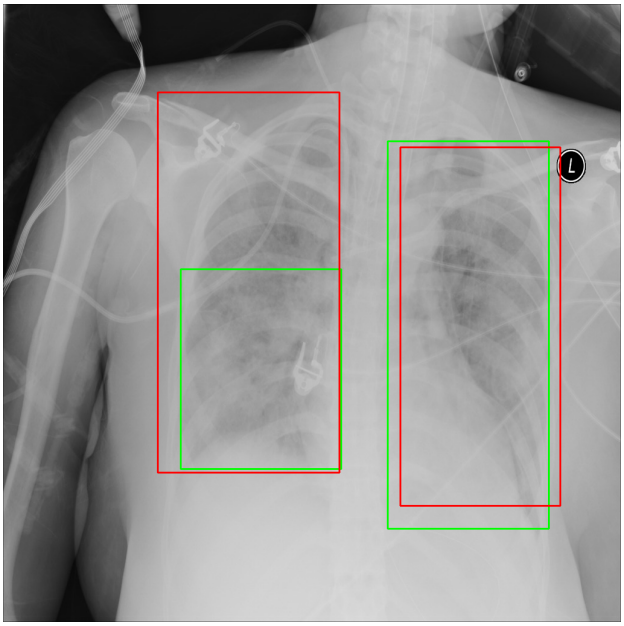

Fig. 8: YOLOv8n example output (case B): miss match in ground-truth (green) vs. prediction (red).

2) Regression heads are insufficient for precise localization. Despite strong classification performance, direct bounding-box regression achieved mean IoU values ≤ 0.21, underscoring the limitations of weakly supervised localization.
3) Multi-level architectures improve spatial accuracy. The YOLOv8n detector, combined with Inception-based classification, achieved superior bounding-box metrics, highlighting the benefit of separating detection from diagnosis in a two-stage design.

Overall, this proof-of-concept demonstrates that multimodal classification paired with task-specific detection offers a practical and interpretable pathway toward automated pneumonia localization. While absolute detection accuracy remains modest compared to fully supervised detectors, the approach balances efficiency and clinical utility, and can be extended to larger backbones, longer training, and stronger annotation schemes for improved performance.

VI. Conclusion

This paper presented a proof-of-concept diagnostic system for pneumonia detection and localization from chest radiographs using multimodal deep learning. Radiographs were combined with patient metadata in fusion classifiers based on ResNet-50, DenseNet-121, and Inception-v3, achieving strong case-level performance with AUCs above 0.88. However, regression-based bounding-box localization yielded limited accuracy (mean IoU $\leq$ 0.21), highlighting the weakness of weakly supervised approaches. To address this, a multi-level pipeline was introduced in which the best multimodal classifier (Inception-v3) gated positive cases, and a YOLOv8n detector was applied for bounding-box prediction. The detector achieved substantially better localization, with mAP@0.5 = 0.38 and mean IoU $\approx$ 0.30, demonstrating the advantage of separating detection from classification.

These findings suggest that multimodal learning is promising for pneumonia diagnosis, and that multi-level architectures provide a practical pathway toward accurate and interpretable clinical decision support. While the present results are modest, they establish a foundation for future work and highlight the overall feasibility of the proposed approach. Potential directions include training larger YOLO variants (e.g., YOLOv8s/m), extending to higher input resolutions and longer training schedules, incorporating stronger annotated datasets, and validating performance across multi-institutional cohorts. Such extensions can further improve localization fidelity and support clinical deployment at scale, ensuring that subsequent research can build progressively on these initial outcomes. As the methodology matures, the same framework can be incrementally refined without altering its core principles. This gradual refinement allows the system to evolve organically while maintaining conceptual consistency. Moreover, the expanded experimentation simply broadens the scope of evaluation rather than shifting the underlying intent. Consequently, the work remains aligned with its original objectives, serving primarily as a stepping stone for more comprehensive studies. Overall, the extended discussion only reinforces the continuity and trajectory of the research without introducing new interpretations.

References

[1] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2016/html/He_Deep_Residual_Learning_CVPR_2016_paper.html

[2] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 2818–2826. [Online]. Available: https://www.cv-foundation.org/openaccess/content_cvpr_2016/papers/Szegedy_Rethinking_the_Inception_CVPR_2016_paper.pdf

[3] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2261–2269. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2017/html/Huang_Densely_Connected_Convolutional_CVPR_2017_paper.html

[4] M. Tan and Q. V. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *Proceedings of the 36th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 97, 2019, pp. 6105–6114. [Online]. Available: https://proceedings.mlr.press/v97/tan19a/tan19a.pdf

[5] P. Rajpurkar, J. Irvin, K. Zhu, B. Yang, H. Mehta, T. Duan, D. Ding, A. Bagul, C. Langlotz, K. Shpanskaya, M. P. Lungren, and A. Y. Ng, "Chexnet: Radiologist-level pneumonia detection on chest x-rays with deep learning," *arXiv preprint arXiv:1711.05225*, 2017. [Online]. Available: https://arxiv.org/abs/1711.05225

[6] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers, "Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 3462–3471. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2017/papers/Wang_ChestX-ray8_Hospital-Scale_Chest_CVPR_2017_paper.pdf

[7] J. Irvin, P. Rajpurkar, M. Ko, Y. Yu, S. Ciurea-Ilcus, C. Chute, H. Marklund, B. Haghgoo, R. Ball, K. Shpanskaya, J. Seekins, D. A. Mong, S. S. Halabi, J. K. Sandberg, R. Jones, D. B. Larson, C. P. Langlotz, B. N. Patel, M. P. Lungren, and A. Y. Ng, "Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, 2019, pp. 590–597. [Online]. Available: https://cdn.aaai.org/ojs/3834/3834-13-6892-1-10-20190701.pdf

[8] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626. [Online]. Available: https://openaccess.thecvf.com/content_iccv_2017/html/Selvaraju_Grad-CAM_Visual_Explanations_ICCV_2017_paper.html

[9] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 2921–2929. [Online]. Available: https://cnnlocalization.csail.mit.edu/Zhou_Learning_Deep_Features_CVPR_2016_paper.pdf

[10] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, ser. LNCS. Springer, 2015, vol. 9351, pp. 234–241. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-319-24574-4_28

[11] D. S. Kermany *et al.*, "Identifying medical diagnoses and treatable diseases by image-based deep learning," *Cell*, vol. 172, no. 5, pp. 1122–1131.e9, 2018. [Online]. Available: https://data.mendeley.com/datasets/rscbjbr9sj/2

[12] Radiological Society of North America, "Rsna pneumonia detection challenge," https://www.kaggle.com/c/rsna-pneumonia-detection-challenge, 2018.

[13] P. Rajpurkar, J. Irvin, A. Bagul, D. Ding, T. Duan, H. Mehta, B. Yang, K. Zhu, D. Laird, R. L. Ball, C. Langlotz, K. Shpanskaya, M. P. Lungren, and A. Y. Ng, "Mura: Large dataset for abnormality detection in musculoskeletal radiographs," *arXiv preprint arXiv:1712.06957*, 2018. [Online]. Available: https://arxiv.org/abs/1712.06957

[14] R. Lindsey, A. Daluiski, S. Chopra, J. Forrester, D. Hanel, J. Herring, E. Gausden, H. Potter, R. Hotchkiss, R. A. Feldman, M. J. Gardner, D. Scheinberg, L. Band, G. An, A. Gupta, K. I. Gruson, M. P. Recht, Y. Fleissig, A. Kots, M. D. Kohli, M. Kohli, and et al., "Deep neural network improves fracture detection by clinicians," *Proceedings of the National Academy of Sciences (PNAS)*, vol. 115, no. 45, pp. 11 591–11 596, 2018. [Online]. Available: https://www.pnas.org/doi/10.1073/pnas.1806905115

[15] J. Olczak, N. Fahlberg, A. Maki, N. Razavian, A. Jilert, A. Stark, O. Sköldenberg, and M. Gordon, "Artificial intelligence for analyzing orthopedic trauma radiographs: Deep learning algorithms—are they on par with humans for diagnosing fractures?" *Acta Orthopaedica*, vol. 88, no. 6, pp. 581–586, 2017. [Online]. Available: https://actaorthop.org/actao/article/view/9748

[16] M. A. Badgeley, J. R. Zech, L. Oakden-Rayner, B. S. Glicksberg, M. Liu, W. Gale, M. V. McConnell, B. Percha, M. P. Snyder, and J. T. Dudley, "Deep learning predicts hip fracture using confounding patient and healthcare variables," *npj Digital Medicine*, vol. 2, no. 31, 2019. [Online]. Available: https://www.nature.com/articles/s41746-019-0105-1

[17] C.-T. Cheng, T.-Y. Ho, T.-Y. Lee, C.-C. Chang, and et al., "Application of a deep learning algorithm for detection and visualization of hip fractures on plain pelvic radiographs," *European Radiology*, vol. 29, no. 10, pp. 5469–5477, 2019. [Online]. Available: https://pubmed.ncbi.nlm.nih.gov/30937588/

[18] J. D. Krogue, K. V. Cheng, K. M. Hwang, P. Toogood, and et al., "Automatic hip fracture identification and functional subclassification with deep learning," *Radiology: Artificial Intelligence*, vol. 2, no. 2, p. e190023, 2020. [Online]. Available: https://pubs.rsna.org/doi/abs/10.1148/ryai.2020190023

[19] M. W. Kim, J. H. Park, S.-Y. Lee, and et al., "Application of convolutional neural networks for distal radius fracture detection in the emergency department," *Clinical and Experimental Emergency Medicine*, vol. 8, no. 2, pp. 120–127, 2021. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC8273672/

[20] A. M. Raisuddin, M. Umer, T. Qaiser *et al.*, "Critical evaluation of deep neural networks for wrist fracture detection (deepwrist)," *Scientific Reports*, vol. 11, no. 1, pp. 1–13, 2021. [Online]. Available: https://www.nature.com/articles/s41598-021-85570-2

[21] W. Gale, L. Oakden-Rayner, G. Carneiro, A. P. Bradley, and L. J. Palmer, "Detecting hip fractures with radiologist-level performance using deep neural networks," in *arXiv preprint arXiv:1711.06504*, 2017. [Online]. Available: https://arxiv.org/abs/1711.06504

[22] G. Kitamura, F. Lareyre, S. Black, and et al., "Deep learning evaluation of pelvic radiographs for position, hardware, and fracture detection," *European Journal of Radiology*, vol. 125, p. 108892, 2020. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC7483754/

[23] V. Hansen *et al.*, "Deep learning performance compared to clinicians for wrist fracture detection: A meta-analysis," *Clinical Radiology*, vol. 79, no. 10, pp. e603–e615, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0720048X24001153